Physiology of Blood & Leukocytes: Set 3

1. During an acute bacterial infection, a significant shift toward immature neutrophils (such as band cells) is observed in the blood peripheral smear. This clinical "shift to the left" conceptually reflects:
a. The destruction of the vascular endothelial cell lining
b. A systemic failure of mitochondrial oxidative phosphorylation
c. Active splenic macrophage clearance of mature red blood cells
d. The bone marrow rapidly releasing its stored pool of developing granulocytes to meet high tissue demand
e. A localized failure of the myenteric nerve plexus
2. Tissue macrophages located within the chronic sinusoids of the liver are highly specialized to filter out enteric pathogens arriving from the gastrointestinal tract. These resident cells are:
a. Microglia
b. Histiocytes
c. Langerhans cells
d. Kupffer cells
e. Alveolar dust cells
3. Conceptually, what is the core physiological mechanism that allows a tissue macrophage to survive for months within a chronic inflammatory zone, whereas a neutrophil dies within hours of executing phagocytosis?
a. Macrophages possess zero lysosomes, preventing self-digestion
b. Macrophages continuously regenerate their internal organelles and lysosomes via active protein synthesis, unlike short-lived, terminally differentiated neutrophils
c. Macrophages completely rely on anaerobic glycolysis for all metabolic functions
d. Macrophages are enclosed in a thick layer of calcified hyaline cartilage
e. Macrophages systematically convert all entering oxygen into hemoglobin
4. A patient presents with a genetic mutation that renders their leukocyte selectin and integrin receptors entirely non-functional. The primary physiological consequence of this structural defect will be:
a. Widespread intravascular coagulation of blood
b. Accelerated maturation of macrocytic megaloblasts
c. Hypersecretion of renal erythropoietin hormone
d. Severe immunodeficiency due to the absolute inability of leukocytes to adhere to endothelium and migrate into injured tissues
e. Total atrophy of the intrinsic laryngeal muscles
5. How does the generation of the phagolysosome conceptually protect the surrounding healthy host tissue during the destruction of an engulfed bacterial cell?
a. It triggers the immediate release of prostacyclin from healthy endothelial cells
b. It converts all released bacterial toxins into soluble plasma albumin
c. It completely halts the filtration of fluid across the capillary wall
d. It isolates the lethal hydrolytic enzymes and reactive oxygen species inside a sealed intracellular vesicle, preventing them from leaking into the extracellular space
e. It shifts the local oxygen-hemoglobin dissociation curve to the left
6. During the early stages of severe tissue trauma, why does the absolute rate of neutrophil migration into the area accelerate dramatically within minutes?
a. Because local tissue macrophages immediately undergo rapid mitotic division

b. Because pre-existing inflammatory mediators like histamine and leukotrienes rapidly increase local capillary pore size and upregulate cell adhesion molecules
c. Because the bone marrow completely ceases all forms of leukopoiesis
d. Because the liver stops synthesizing plasma clotting factors
e. Because the neutrophils lose their internal nuclei to become smaller fragments

7. In an individual demonstrating an inherited absence of the enzyme myeloperoxidase inside their neutrophilic granules, which specific intracellular capability is pathologically lost?

a. The ability to migrate via ameboid motion toward a chemotactic gradient
b. The capability to engulf bacteria via pseudopod formation
c. The ability to convert hydrogen peroxide into highly lethal hypochlorite ions during the respiratory burst
d. The synthesis of Class II Major Histocompatibility Complex proteins
e. The structural retraction of loose fibrin meshes

8. Conceptually, why does the administration of high doses of synthetic glucocorticoids (corticosteroids) systematically suppress the body's natural inflammatory response?

a. Because they directly accelerate the rate of red cell hemolysis
b. Because they trigger the immediate degranulation of all tissue mast cells
c. Because they stabilize lysosomal membranes and inhibit phospholipase A2, markedly reducing the synthesis of inflammatory leukotrienes and prostaglandins
d. Because they double the concentration of plasma fibrinogen
e. Because they block the passage of lymph through the thoracic duct

9. The clonal selection theory states that every single lymphocyte clone is genetically pre-programmed to recognize one specific antigenic determinant. This extreme specificity is structurally determined by:

a. The continuous rearrangement of constant region genes in the liver
b. The specific sequence of amino acids in the hypervariable loops of the variable regions of its surface receptors or antibodies
c. The presence of a biconcave profile optimized for rapid gas diffusion
d. The thickness of the surrounding loose connective tissue capsule
e. The velocity of fluid movement through the peritubular capillaries

10. When an antigen-presenting cell (APC) phagocytoses a viral particle, it degrades the viral proteins into small peptide fragments. To present these fragments to a helper T cell, the APC must bind them to:

a. Class I Major Histocompatibility Complex (MHC) proteins
b. Class II Major Histocompatibility Complex (MHC) proteins
c. Preformed monomeric IgE antibodies
d. Endothelial selectin receptors
e. Membrane attack complexes

11. Conceptually, what is the fundamental functional difference between the presentation of an antigen on Class I MHC versus Class II MHC proteins?

a. Class I MHC presents only lipid molecules, while Class II presents carbohydrates
b. Class I MHC is only found on red blood cells, while Class II is on white blood cells
c. Class I MHC presents endogenous antigens to cytotoxic T cells (CD8+) to signal host cell destruction, while Class II presents exogenous antigens to helper T cells (CD4+) to activate the wider adaptive response

d. Class I MHC activates the intrinsic clotting pathway, while Class II activates the extrinsic pathway
e. Class I MHC triggers immediate histamine release, while Class II triggers hepcidin release

12. Following the primary exposure to a foreign antigen, a characteristically long lag phase of several days occurs before significant antibody titers appear in the blood. This delayed onset conceptually represents:

a. The time required for the spleen to clear all old erythrocytes
b. The structural transformation of neutrophils into tissue macrophages
c. The time necessary for rare specific lymphocyte clones to encounter the antigen, undergo multiple rounds of clonal proliferation, and differentiate into mature plasma cells
d. The slow conjugation of free bilirubin inside hepatic cords
e. The continuous active breakdown of pulmonary surfactant

13. A patient with a rare genetic defect lacks the constant region (Fc portion) of their circulating IgG antibodies but possesses fully functional variable regions. Which antibody-mediated function remains normal in this patient?

a. The direct chemical binding and neutralization of specific bacterial toxins
b. The activation of the C1 component of the complement cascade
c. The attachment of the antibody to the Fc receptors of phagocytic cells for opsonization
d. The binding of the antibody to the surface membranes of tissue mast cells
e. The transport of the antibody across the placental barrier to the fetus

14. The classical pathway of the complement system is activated by antigen-antibody complexes. In contrast, the alternative pathway of complement activation is triggered directly by:

a. High levels of circulating plasma erythropoietin hormone
b. The contact of complement proteins with specific polysaccharide molecules on the surface membranes of invading microorganisms without antibodies
c. The release of Tissue Factor from damaged visceral organs
d. The direct binding of Vitamin K to plasma apoferritin
e. The myogenic spasm of damaged large systemic arteries

15. The complement system generates a terminal membrane attack complex (MAC). The biophysical mechanism by which the MAC physically kills an invading bacterial cell is by:

a. Enzymatically digesting the bacterial nuclear genome
b. Converting the bacterial internal cytoplasm into a rigid fibrin gel
c. Constructing large, open hydrophilic channels across the bacterial lipid bilayer, causing rapid water influx, osmotic swelling, and lysis
d. Blocking the bacterial absorption of dietary iron salts
e. Activating the involuntary smooth muscle layers of the host tissue

16. Conceptually, how do helper T cells (CD4+) enhance the functional destructive capabilities of tissue macrophages during a cell-mediated immune response?

a. They convert macrophages into non-nucleated platelet fragments
b. They secrete specific lymphokines, primarily interferon-gamma, which powerfully activate macrophages to maximize their intracellular lysosomal killing capacity
c. They completely strip macrophages of all Class II MHC surface proteins
d. They synthesize a dense layer of plasma albumin around the macrophage cell wall
e. They trigger the immediate apoptosis of all tissue macrophages

17. Cytotoxic T cells (CD8+) use surface receptors to scan host cells. When they identify a host cell presenting a foreign viral peptide on a Class I MHC protein, they destroy that cell to:
a. Stimulate the bone marrow to accelerate erythropoiesis
b. Eliminate the cellular factory where the virus is actively replicating, halting the spread of the infection
c. Free up space for the immediate deposition of a loose platelet plug
d. Conjugate free bilirubin into water-soluble compounds
e. Induce local smooth muscle contraction via the myenteric plexus
18. Memory B and T lymphocytes remain dormant in lymphoid tissues for decades after the primary infection has cleared. These memory cells differ conceptually from naive lymphocytes because:
a. They completely lack a cell nucleus and mitochondria
b. They exist in significantly larger numbers, are highly sensitive to lower concentrations of the antigen, and execute immediate, explosive differentiation upon re-exposure
c. They are completely confined within the bone marrow stroma
d. They express preformed membrane attack complexes on their outer surfaces
e. They rely entirely on the absorption of gastric intrinsic factor
19. During an immediate Type I hypersensitivity reaction, why does the localized or systemic smooth muscle contraction and vascular leakage occur within minutes of allergen re-exposure?
a. Because the allergen directly degrades the circulating plasma fibrinogen
b. Because pre-sensitized mast cells and basophils are already coated with allergen-specific IgE, causing instantaneous degranulation and release of histamine upon binding the allergen
c. Because cytotoxic T cells execute widespread delayed destruction of host vessels
d. Because the celiac trunk artery undergoes an acute structural spasm
e. Because the complement system fails to clear circulating immune complexes
20. A patient presents with severe autoimmune hemolytic anemia, where their immune system synthesizes antibodies against their own red blood cell surface antigens. This disease process is a classic example of:
a. Type I hypersensitivity
b. Type II hypersensitivity
c. Type III hypersensitivity
d. Type IV hypersensitivity
e. Central immunological tolerance
21. Conceptually, what is the key pathological event that drives tissue damage in a Type III hypersensitivity reaction (such as systemic lupus erythematosus or acute glomerulonephritis)?
a. IgE antibodies triggering explosive degranulation of tissue mast cells
b. Cytotoxic T cells directly lysing healthy vascular endothelial cells
c. Soluble antigen-antibody complexes depositing in vessel walls, activating the complement cascade, and recruiting destructive neutrophils
d. A severe lack of functional Vitamin K halting prothrombin synthesis
e. Moving red cells into the alveolar lumens due to high venous pressure
22. The tuberculin skin test (Mantoux test) is used to detect prior exposure to Mycobacterium tuberculosis. A positive reaction manifests as an induration at the

injection site that reaches peak intensity 48 to 72 hours later. This delay represents the time required for:
a. Specific B cells to differentiate into active plasma cells and release IgM
b. Antigen-specific T cells and circulating monocytes to migrate to the skin site, recognize the antigen, and release inflammatory cytokines
c. The liver to clear all injected bacterial proteins from the blood
d. The complement system to synthesize the terminal membrane attack complex
e. The bone marrow to accelerate the production of mature erythrocytes
23. Immunological tolerance prevents the body's adaptive immune system from destroying its own healthy organs. Central tolerance is established conceptually during early development by:
a. The continuous active tracking of circulating tissue macrophages
b. Generating a high concentration of plasma albumin to shield host proteins
c. Deleting or inactivating self-reactive T and B lymphocyte clones inside the primary lymphoid organs (thymus and bone marrow) when they encounter self-antigens
d. Turning off the expression of all Class I MHC proteins throughout the body
e. Excreting all self-reactive antibodies via the main pancreatic duct
24. Natural Killer (NK) cells identify and destroy malignant tumor cells that have mutated to downregulate or completely stop expressing their Class I MHC proteins. Why are NK cells effective in this specific scenario where cytotoxic T cells are completely powerless?
a. Because NK cells completely lack any cytoplasmic granules or destructive proteins
b. Because cytotoxic T cells require the absolute presence of Class I MHC presentation to identify a target, whereas NK cells are activated by the absolute absence of Class I "self" signals on a host cell
c. Because NK cells can cross the immature placental barrier easily
d. Because NK cells are derived directly from the embryonic yolk sac endoderm
e. Because NK cells synthesize massive quantities of specific IgG antibodies
25. A newborn infant receives passive immunity via the transfer of maternal antibodies across the placental barrier. This temporary protective shield is composed exclusively of antibodies belonging to the class:
a. IgM
b. IgD
c. IgE
d. IgG
e. IgA
26. Conceptually, what is the primary immunological benefit of the structural cross-linking of antigens during the process of antibody-mediated agglutination?
a. It renders the antigen completely soluble in pure host lipids
b. It blocks the activation of the terminal membrane attack complex
c. It binds multiple separate antigenic particles into a massive, immobile molecular cluster, limiting its spread and making it a highly attractive target for phagocytosis
d. It converts the antigen into a functional clotting factor look-alike
e. It triggers the immediate secretion of bicarbonate-rich duodenal fluid
27. In a patient demonstrating severe, unmanaged chronic inflammation, high levels of systemic interleukin-6 (IL-6) travel to the liver. This signaling pathway induces hepatocytes to dramatically upregulate the synthesis of:
a. Bile salts and bilirubin glucuronide

b. Acute-phase reactants (such as C-reactive protein and fibrinogen)
c. Gastric intrinsic factor and pepsinogen
d. Thyroid and cricoid branchial cartilages
e. Erythropoietin and thrombopoietin receptors

28. Why does the human body utilize a specialized lymphatic drainage system to transport long-chain dietary fats absorbed from the intestinal villi into the blood, rather than draining them directly into the portal vein?

a. Because fats are too large to pass through the tightly sealed endothelial junctions of the portal capillaries, requiring the wide, highly permeable pores of the lymphatic lacteals
b. Because fats must be exposed to central thymic processing first
c. Because the portal vein completely lacks any internal fluid movement
d. Because intestinal fats must be converted into plasma albumin molecules
e. Because the celiac trunk strictly repels all lipophilic compounds

29. Conceptually, the process of somatic hypermutation occurring within the germinal centers of a lymph node during an active infection ensures that the responding B cells undergo:

a. A total loss of their cell nuclei to become platelets
b. Affinity maturation, selecting for B-cell clones that express surface antibodies with progressively higher binding affinities for the target antigen
c. A complete regression into pluripotent hematopoietic stem cells
d. Sudden conversion into intrinsic laryngeal muscle fibers
e. Accelerated synthesis of fetal hemoglobin HbF chains

30. The absolute master switch that coordinates the balance between the humoral immune response (B cells) and the cell-mediated immune response (T cells) is directed by the differentiation of naive T helper cells (Th0) into either:

a. Type I or Type II pneumocytes
b. Alpha or beta globulin plasma fractions
c. Th1 cells (driving cell-mediated defense) or Th2 cells (driving humoral antibody defense)
d. Chief cells or parietal cells of the gastric mucosa
e. Granulocytic neutrophils or agranulocytic monocytes

31. When a cytotoxic T cell (CD8+) delivers its lethal hit to a virus-infected cell, the injected granzymes activate a cascade of intracellular host enzymes known as:

a. Caspases (which initiate programmed cell death or apoptosis)
b. Carbonic anhydrase polymers
c. Proteases of the clotting cascade Stage 1
d. Cytochrome P450 detoxifying enzymes
e. Cyclooxygenase units that synthesize thromboxane

32. Conceptually, how does the physiological presence of Regulatory T cells (Tregs) suppress the development of systemic autoimmune diseases?

a. They synthesize vast quantities of preformed membrane attack complexes
b. They actively secrete inhibitory cytokines like IL-10 and TGF-beta, which turn off the activation of self-reactive T cells in peripheral tissues
c. They accelerate the fragmentation of bone marrow megakaryocytes
d. They consume all circulating plasma transferrin molecules
e. They systematically destroy the Class II MHC proteins on dendritic cells

33. An individual undergoes a surgical splenectomy following a severe traumatic injury. Conceptually, this patient will experience a lifelong increased risk of:

a. Severe macrocytic megaloblastic anemia

b. A complete shutdown of renal erythropoietin synthesis
c. Fulminant systemic infections by encapsulated bacteria due to a loss of the spleen's primary blood-filtering and antibody-producing lymphoid tissue
d. Total failure of localized vascular smooth muscle spasms
e. Hyper-absorption of fat-soluble dietary vitamins

34. Why do professional antigen-presenting cells express costimulatory proteins (such as B7) on their surface membranes only after they have encountered an active pathogen?

a. To accelerate the systemic breakdown of old host red cells
b. To ensure that naive T cells are activated only when a real infection is present, preventing accidental activation by harmless self-antigens
c. To increase the thickness of the loose connective tissue lamina propria
d. To convert soluble fibrinogen into a stable intravascular thrombus
e. To fix the APC securely to the posterior trachealis smooth muscle

35. A patient with severe, uncontrolled type I diabetes presents with a total lack of insulin. This hormone deficiency indirectly impairs the acquired immune system because leukocytes require insulin to effectively execute:

a. The chemical synthesis of hemoglobin polypeptide chains
b. The direct active absorption of dietary iron salts in the jejunum
c. Intracellular glucose uptake to generate the high metabolic energy needed for rapid clonal proliferation and migration
d. The vertical growth of the embryonic tracheoesophageal septum
e. The expression of preformed anti-D Rh antibodies in the plasma

36. Conceptually, how does the human body ensure that the acquired immune system remains highly localized to an infection zone rather than causing systemic inflammation?

a. Lymphokines have exceptionally short half-lives in the blood and act primarily as local paracrine signals within the lymphoid tissues and infection sites
b. All activated T and B lymphocytes are permanently locked inside the spleen
c. Antibodies can only circulate within the lymphatic vessels, never entering blood
d. Circulating plasma albumin actively degrades all specific immunoglobulins
e. Endothelial cell tight junctions permanently seal all systemic organs

37. In an individual demonstrating an inherited deficiency of the complement regulatory protein Decay-Accelerating Factor (DAF) on their red blood cell membranes, the complement system will pathologically:

a. Fail to initiate the classical pathway entirely
b. Execute uncontrolled, spontaneous assembly of the membrane attack complex on the host's own erythrocytes, causing intravascular hemolytic anemia
c. Completely stop the chemotaxis of neutrophils to infection sites
d. Accelerate the rate-limiting step of prothrombin activator synthesis
e. Convert the erythrocyte shape into a rigid, non-functional sickle cell

38. The severe mucosal swelling and profound watery secretions characterizing an atopic individual's response to airborne pollen are conceptually driven by histamine acting locally to:

a. Cause severe myogenic contraction of the large conducting systemic arteries
b. Induce localized arteriolar vasodilation and downregulate endothelial tight junctions, causing massive plasma exudation into the respiratory mucosa
c. Suppress the migration of neutrophils through the capillary pores

d. Synthesize a dense layer of hyaline cartilage around the nasal cavity
e. Inhibit the exocrine secretions of the sub submucosal Brunner's glands

39. Widespread, systemic activation of the complement cascade by circulating bacterial endotoxins during Gram-negative septicemia poses a fatal hazard because it triggers:

a. Widespread bone marrow aplasia and a total loss of pluripotent cells
b. Massive, systemic release of anaphylatoxins (C3a and C5a), causing profound, unmanageable vasodilation and distributive shock
c. High-pressure accumulation of lipid-soluble bilirubin in the basal ganglia
d. The total destruction of the gastric intrinsic factor pathway
e. Spontaneous calcification of the intrinsic laryngeal muscle cores

40. The structural mechanism of central T-cell tolerance inside the thymic cortex involves a selection phase where developing T cells that express receptors with a high binding affinity for self-peptide-MHC complexes are induced to undergo:

a. Rapid, uncontrolled clonal proliferation
b. Transformation into phagocytic tissue macrophages
c. Apoptosis (clonal deletion via negative selection)
d. Extrusion of their cell nuclei to enter circulation
e. Continuous active secretion of pentameric IgM antibodies

Answer Key Table (Set 3)

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40
d	b	b	b	c
d	c	b	c	b
b	c	b	b	c
d	a	b	a	a
d	b	c	b	c
b	c	c	b	a
c	b	c	a	b
c	b	b	c	c